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Research Overview

My research centers on multimodal foundation models, with key areas of focus including:

- **Generative Modeling for Scientific Discovery**, developing diffusion and flow-matching methods for general, medical, and biological generation, with an emphasis on faithful structure and data efficiency.
- **Self-Evolving Long-Horizon Agents**, developing agents that continually learn from experience and refine their planning, reasoning, memory, and tool-use capabilities across extended tasks and interactions.
- **Pre-training and Post-training of MLLMs**, improving domain adaptation, spatial reasoning, and robust alignment through data mixture design, instruction tuning, and preference optimization.

Education

Ph.D. Engineering and IT Sept. 2026 – Present	The University of Melbourne Advisor: Jianzhong Qi
M.E. Computer Science and Technology Sept. 2023 – Mar. 2026	Tongji University Advisor: Guosun Zeng
B.E. Intelligence Science and Technology Sept. 2019 – Jun. 2023	Dalian Maritime University

Selected Publications

The * sign denotes equal contribution. The † sign denotes corresponding author.

Peer-Reviewed Papers

- C8 Yuqi Li*, Kuiye Ding*, Chuanguang Yang†, Haoxuan Wang, Huiran Duan, Junming Liu, et al. **DDTime: Dataset Distillation with Spectral Alignment and Information Bottleneck for Time-Series Forecasting**. In *Conference on Neural Information Processing Systems (NeurIPS)*. 2026.
- J3 Junming Liu, Yanting Gao, Yuqi Li, et al. **Mosaic: Data-Free Knowledge Distillation via Mixture-of-Experts for Heterogeneous Distributed Environments**. In *Knowledge-Based Systems*. 2026.
- J2 Yuqi Li*, Junhao Dong*, Jiao Liu, Piotr Koniusz, Hansheng Zeng, Chuanguang Yang, Junming Liu, et al. **Evolving Multimodal Models for Physical Dynamics: A Multi-objective Neuroevolution Approach**. In *IEEE Transactions on Evolutionary Computation*. 2026.

- C7 Yuqi Li*, Siwei Meng*, Chuanguang Yang, Weilun Feng, Junming Liu, et al. **A Comprehensive Survey of Interaction Techniques in 3D Scene Generation**. In *International Joint Conference on Artificial Intelligence (IJCAI-ECAI)*. 2026.
- C6 Junming Liu, Yusen Zhang, Rongchao Zhang, et al. **Domain-Adaptive Model Merging Across Disconnected Modes**. In *IEEE International Conference on Acoustics, Speech, and Signal Processing (ICASSP)*. 2026.
- C5 Aoqi Wu, Junming Liu, Yuwei Zhang, et al. **AMID: Model-Agnostic Dataset Distillation by Adversarial Mutual Information Minimization**. In *The ACM Web Conference (WWW)*. 2026.
- C4 Junming Liu, Yifei Sun, Weihuang Cheng, et al. **ReBrain: Brain MRI Reconstruction from Sparse CT Slice via Retrieval-Augmented Diffusion**. In *IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*. 2026.
- C3 Yanting Gao, Yepeng Liu, Junming Liu, et al. **Boosting Adversarial Transferability via Commonality-Oriented Gradient Optimization**. In *Chinese Conference on Pattern Recognition and Computer Vision (PRCV)*. 2025.
- C2 Pei Liu, Xin Liu, Ruoyu Yao, Junming Liu, et al. **HM-RAG: Hierarchical Multi-Agent Multimodal Retrieval Augmented Generation**. In *ACM International Conference on Multimedia (ACM MM)*. 2025.
- C1 Junming Liu, Siyuan Meng, Yanting Gao, et al. **Aligning Vision to Language: Annotation-Free Multimodal Knowledge Graph Construction for Enhanced LLM Reasoning**. In *IEEE/CVF International Conference on Computer Vision (ICCV)*. 2025.
- J1 Yujin Kang*, Junming Liu*, Haiyan Cui†. **AI-Driven Assessment of Lip Volume Improvement Using Hyaluronic Acid Fillers: A Comprehensive Analysis**. In *Aesthetic Plastic Surgery*. 2025.

Preprints and Under Review

- p11 Junming Liu*, Yifei Sun*, Weihua Cheng, et al. **MemVerse: Multimodal Memory for Lifelong Learning Agents**. In *Conference of the European Chapter of the Association for Computational Linguistics (EACL)*. 2027.
- P10 Junming Liu, Haodong Lei, Maonan Wang, et al. **Mem-O: Learning to Orchestrate Memories**. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2027. Under Review.
- P9 Haodong Lei, Junming Liu, Yirong Chen, et al. **MemCoT: Test-Time Scaling through Memory-Driven Chain-of-Thought**. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2027. Under Review.
- P8 Haodong Lei, Junming Liu, Yirong Chen, et al. **TransMem: Transforming Hidden States into Memory for Large Language Models**. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2027. Under Review.
- P7 Yuqi Li*, Junming Liu*, Heejeong Nam, et al. **Teacher-Guided LeJEPa: Semantic Distillation with Independent Geometry Control**. In *AAAI Conference on Artificial Intelligence (AAAI)*. 2027. Under Review.
- P6 Junming Liu, Yifei Sun, Weihua Cheng, et al. **Hierarchical Memory Orchestration for Personalized Persistent Agents**. In *ACM Conference on Intelligent User Interfaces (IUI)*. 2027. Under Review.
- P5 Siyuan Meng*, Junming Liu*, Yirong Chen, et al. **MemSelector: Plug-and-Play Dynamic Evidence Selection for Memory-Grounded Reasoning**. In *ACM International Conference on Web Search and Data Mining (WSDM)*. 2027. Under Review.
- P4 Junming Liu, Yuqi Li, Yifei Sun, et al. **Self-Evolving Spatial Reasoning in Vision Language Models via Geometric Logic Consistency**. In *International Conference on Learning Representations (ICLR)*. 2027. Under Review.
- P3 Rongchao Zhang, Junming Liu, Haoran Liu, et al. **Be an Artist Who Revises: Learning Fluorescence Microscopy Characterization with Non-Instantaneous Flow Matching**. In *Conference on Neural Information Processing Systems (NeurIPS)*. 2026. Under Review.
- P2 Junming Liu*, Yuqi Li*, Shiping Wen, et al. **Hit-RAG: Learning to Reason with Long Contexts via Preference Alignment**. In *IEEE Transactions on Artificial Intelligence*. Under Review.

P1 Junming Liu, Yuqi Li, Yanting Gao, et al. **FedRecon: Missing Modality Reconstruction in Heterogeneous Distributed Environments**. In *IEEE Transactions on Multimedia*. Under Review.

Honors and Awards

National College Mathematics Competition, National First Prize (1%) Chinese Mathematical Society	Dec. 2022
Excellent Student Scholarship (10%) Dalian Maritime University	Sept. 2021, 2022
Leadership and Communication Scholarship Dalian Maritime University	Sept. 2020

Research Experience

- Recursive Self-Improvement for Foundation Models** **Alibaba Tongyi Lab**
Advisor: Dr. Yingda Chen *Aug. 2026 – Present*
 - Developed end-to-end pre-training and post-training pipelines for foundation models with self-evolving long-context reasoning.
 - Built and open-sourced an RSI framework with supporting models on ModelScope, enabling reproducible self-evolution research for the community.
- On-Policy Distillation for World Models** **Mohamed bin Zayed University of Artificial Intelligence**
Advisor: Dr. Dongxia Wu *Aug. 2026 – Present*
 - Studied on-policy distillation methods for world models, focusing on transferring privileged temporal knowledge into deployable student views.
 - Explored identifiability-aware training objectives to avoid distilling information that is not recoverable from the student observation.
- Reasoning in Multimodal Large Language Models** **Shanghai AI Laboratory**
Advisor: Dr. Ding Wang *Jan. 2025 – Aug. 2026*
 - Studied reasoning, memory, and world modeling in multimodal foundation models, with emphasis on reliable long-context and spatially grounded inference.
 - Developed methods for medical-domain mid-training, spatial-reasoning post-training, retrieval-augmented generation, and memory-augmented agents for long-horizon interaction.
- Multimodal Generative Modeling for Cellular Perturbation** **Stanford University**
Advisor: Dr. Yuhui Zhang *Apr. 2026 – Jul. 2026*
 - Studied generative modeling for cellular perturbation, connecting microscopy-derived morphology with gene-expression responses under drug interventions.
 - Developed flow-matching models for faithful phenotype mapping, aiming to provide data-efficient generative baselines for computational pharmacology.
- Vision-Language Models for Medical Imaging** **Tongji University**
Advisor: Prof. Guosun Zeng *Sept. 2023 – Mar. 2026*
 - Worked on brain medical imaging with diffusion models and VLMs, including modality reconstruction, abnormality detection, and segmentation.

- Explored privacy-preserving distributed training for lightweight model collaboration across heterogeneous medical data and Non-IID clinical settings.

- **Text Sentiment Analysis Based on BERT Model**
Advisor: Prof. Yijia Zhang

Dalian Maritime University
Sept. 2021 – May 2022

- Integrated BERT with LDA topic modeling for social-media sentiment analysis, supporting public-opinion tracking during the COVID-19 pandemic.

Internship Experience

Researcher Aug. 2026 – Present	Tongyi Lab, Token Foundry, Alibaba Conducting research on self-evolving AI systems.
Researcher Jan. 2025 – Aug. 2026	Shanghai AI Laboratory Conducted research on AI agents.
Algorithm Engineer Oct. 2024 – Dec. 2024	Beijing Jinxixin Network Technology Co., Ltd. Constructed and trained vertical-domain large models for maritime law scenarios.
Embedded Engineer Jun. 2023 – Aug. 2023	Shanghai NIO Automobile Co., Ltd. Controlled steering systems using multimodal sensing for rollover prevention in tire blowout scenarios.

Skills

- **Languages:** Chinese (native), English (IELTS: 7.0), Japanese, German.
- **Programming:** C, C++, Python, Java, Go, SQL, Rust, MPI, NCCL, DeepSpeed, Megatron-LM, DDP, FSDP, vLLM, verl, TRL.